House Price Prediction (Week 1)

Kanchan

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics. The project involved data exploration, cleaning, preprocessing, model training, evaluation, visualization, and business insight generation using a real-world housing dataset.

Dataset Overview

The dataset contained 545 housing records with information such as area, number of bedrooms, bathrooms, stories, parking availability, road access, furnishing status, and other amenities. The target variable was house price.

Data Preprocessing

The dataset was inspected for missing values and duplicate records. Categorical variables including main road access, guest room availability, basement availability, hot water heating, air conditioning, preferred area, and furnishing status were converted into numerical form using One-Hot Encoding. All available features were retained because they represented meaningful characteristics that could influence house prices.

Model Development

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into training and testing sets using an 80:20 ratio.

Model Performance

A) Linear Regression

* MAE: 970,043
* RMSE: 1,324,507
* R^2 Score: 0.653

B) Random Forest Regressor

* MAE: 1,013,969
* RMSE: 1,398,116
* R^2 Score: 0.613

Linear Regression achieved better performance across all evaluation metrics and was selected as the best-performing model.

Key Findings

Feature importance analysis indicated that area was the most influential factor affecting house prices. Bathrooms, air conditioning, parking availability, number of stories, and bedrooms also contributed significantly to property valuation.

The price distribution was positively skewed, indicating the presence of a small number of premium-priced houses. Correlation analysis showed positive relationships between price and features such as area, bathrooms, parking, and air conditioning.

Business Recommendation

Real estate businesses should prioritize and highlight high value property features such as larger living area, additional bathrooms, parking facilities, and air conditioning when advertising properties. These attributes have a strong influence on market value and can significantly impact buyer decisions.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. Linear Regression provided reliable predictive performance and revealed important insights into the factors that influence residential property prices.